

The Primer: Your one-liner

Enterprise Safety & Compliance Copilot RAG app helps safety officers and inspectors query enterprise handbooks and inspection procedures to answer compliance and hazard questions in a Streamlit web application with strict zero-hallucination source citations

Corpus:

- Sources & Document Count: The corpus consists of over 400 pages of enterprise safety and compliance documents (handbooks and manuals).
- Formats: Standard Portable Document Format (`.pdf`).
- Language: English.
- Source of Truth: Government regulatory and occupational safety bodies, specifically the Occupational Safety and Health Administration (OSHA) and industrial safety administration standards (comprising publications such as *OSHA 2209*, *OSHA 3885*, *PH25-I-2 Citation and Order Writing Handbook*, and *PH26-V-1 Electrical Inspection Procedures Handbook*).

Ingestion + freshness

- Document Ingestion: Verified PDF handbooks are loaded into the pipeline from the local directory using document loaders.
- Text Extraction & Cleaning:
  Raw text is extracted from the PDF pages.
  Boilerplate headers, footers, and page numbers are stripped to prevent noise from polluting the vector space.
  Special characters and formatting artifacts are normalized, and HTML/XML entities are decoded.
- Chunking: The cleaned text is split into semantic chunks with controlled overlap using text splitters, ensuring context boundaries remain intact before being embedded and indexed into your Chroma and BM25 hybrid retrieval stores.

Chunking+Embedding

- Chunking Strategy & Size: The application uses a fixed-size chunking strategy set to 512 tokens with controlled overlap to segment the cleaned regulatory handbook text. This ensures paragraphs and sections are broken down into digestible windows small enough to fit within model context limits while preserving contiguous local context across dense regulatory rules.
- Semantic vs. Fixed: Fixed-size chunking was chosen over semantic chunking to maintain predictable, uniform boundaries across structured compliance tables, citation rules, and multi-paragraph safety guidelines without missing localized details.
- Embedding Model & Dimensions: The application utilizes OpenAI's text-embedding-3-small model configured with a reduced vector dimension of 384.
- Why: Configuring `text-embedding-3-small` down to 384 dimensions significantly cuts vector storage and indexing memory overhead while retaining robust semantic retrieval performance for low-latency hybrid search across enterprise safety handbooks

Retrieve

- Store Architecture: The application employs a dual-storage setup combining Chroma (for dense vector similarity search) and BM25 (for sparse keyword and term-matching search).
- Dense vs. Sparse vs. Hybrid:
  Dense retrieval uses `text-embedding-3-small` vectors (configured at 384 dimensions) to capture underlying semantic meaning, synonyms, and conceptual intent.
  Sparse retrieval (`BM25`) relies on exact keyword matching, ensuring specific alphanumeric identifiers, standard codes, or regulatory terms (like "OSHA 2209" or specific section numbers) are never missed.
  Hybrid search fuses both approaches using Reciprocal Rank Fusion (RRF), combining the strengths of semantic understanding and exact lexical keyword hits.

- Top-K: The retriever is configured to fetch the top-k most relevant document chunks (e.g., $k = 4$) from the hybrid search pool, ensuring the context window passed to the LLM is concise, highly relevant, and strictly focused on the user's query.

Highlights:

Here are the key production engineering highlights and architectural mechanisms of your Safety & Compliance Copilot that operate behind the scenes outside of the visual diagrams:

1. Rigorous RAG Evaluation (Evals)

While the pipeline handles retrieval and generation, production-grade reliability relies on systematic evaluation across three core dimensions (often referred to as the RAG Triad):

- **Context Precision & Recall:** Evaluating whether the top-k retrieved chunks from Chroma and BM25 actually contain the exact safety guidelines or regulatory clauses required to answer the query.
- **Faithfulness (Groundedness):** Measuring whether the generated response by `gpt-4o-mini` is strictly derived *only*from the retrieved context, ensuring zero ungrounded assertions or hallucinated compliance rules.
- **Answer Relevance:** Assessing whether the final output directly and concisely addresses the safety officer's specific operational question without adding unnecessary conversational fluff.

2. Reciprocal Rank Fusion (RRF) Score Normalization

Instead of relying on a resource-heavy cross-encoder model to re-rank search results, your `HybridRetriever` uses **Reciprocal Rank Fusion**.

- It combines the ordinal rankings from dense vector similarity (Chroma) and lexical keyword matching (BM25) using a rank-scoring formula (typically with a constant k=60).
- This prevents score-scale mismatch issues (where vector distances and BM25 TF-IDF scores have completely different mathematical distributions) and surfaces documents that perform well in *both* semantic intent and exact alphanumeric/statute identification.

3. Dynamic Context-Aware Follow-Up Generation

Beyond static preset prompts, the copilot features a secondary downstream chain dedicated to user intent progression:

- After the primary answer is generated, a lightweight prompt template ingests the active context, the user's question, and the assistant's response.

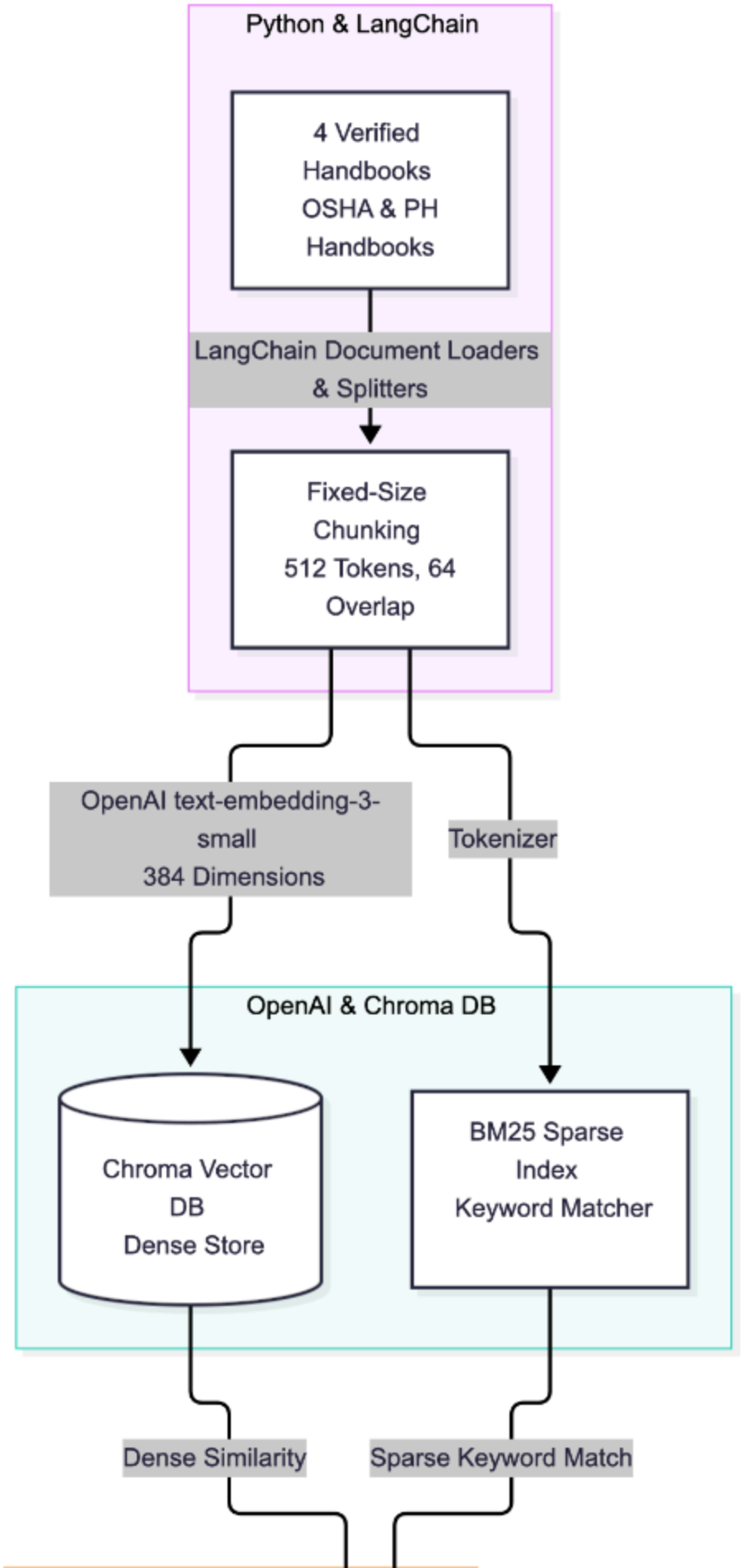
- It dynamically outputs two short, drill-down follow-up questions specific to the regulatory text just retrieved, which are instantly bound to interactive UI buttons (`st.session_state.dynamic_follow_ups`).

4. Zero-Hallucination Prompt Architecture & Guardrails

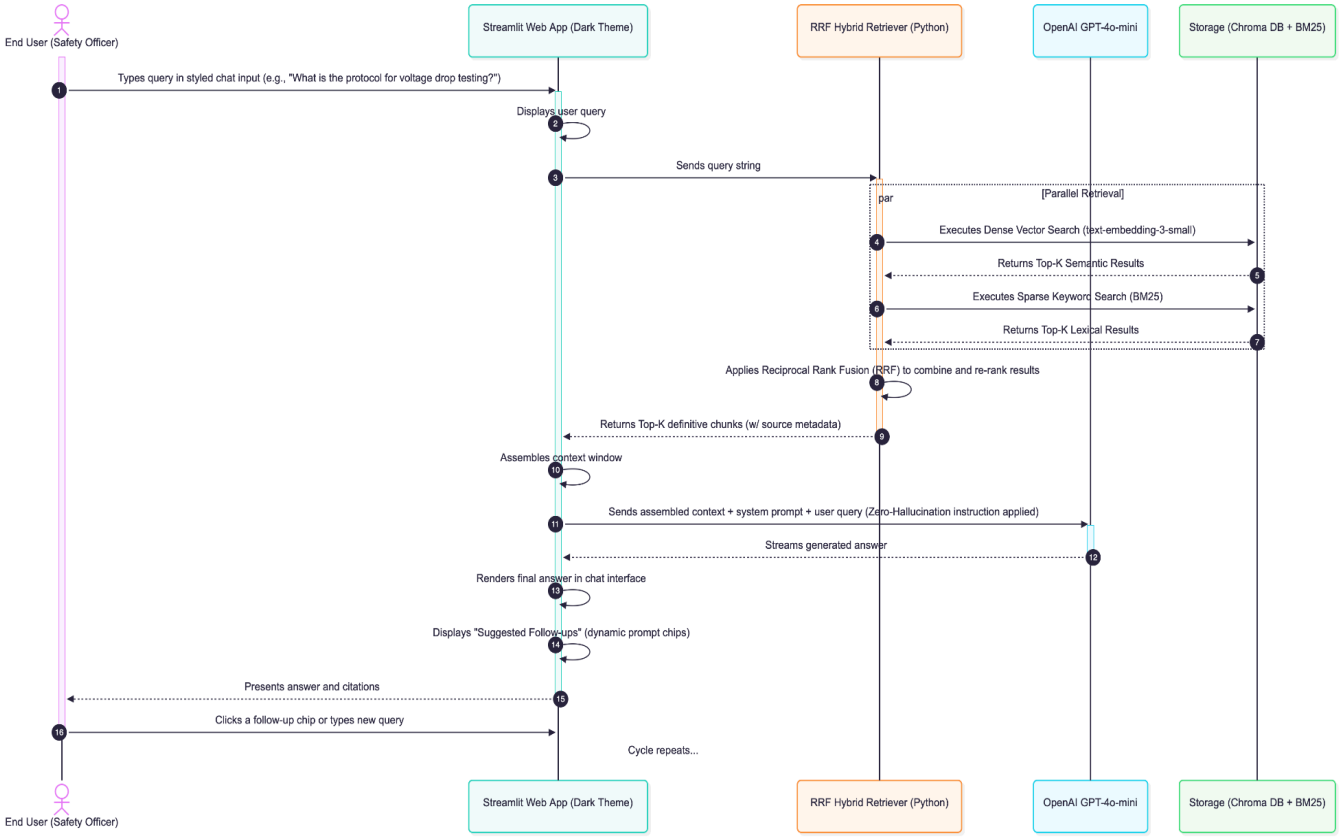
The generation layer is heavily locked down to meet enterprise safety compliance standards:

- **Low Temperature Setting:** `temperature=0.1` minimizes stochastic variation in the LLM's output.
- **Strict Source Binding:** System prompts explicitly instruct the model to fail gracefully or state that information is missing if the context does not contain the answer, coupled with mandatory metadata injection (`source_file` and chunk indexing) displayed inside expandable UI inspectors.

Technical Architecture:


Python & LangChain
4 Verified Handbooks OSHA & PH Handbooks
LangChain Document Loaders & Splitters
Fixed-Size Chunking 512 Tokens, 64 Overlap
OpenAI text-embedding-3-small 384 Dimensions
Tokenizer
OpenAI & Chroma DB
Chroma Vector DB Dense Store
BM25 Sparse Index Keyword Matcher
Dense Similarity
Sparse Keyword Match

Product work flow:



Learnings:

Architecture & Engineering Evolution: Scaling from Minimal Handbooks to Massive Telemetry Datasets

Building the Safety & Compliance Copilot involved a significant architectural scaling journey, moving from a lightweight proof-of-concept to a large-scale, high-performance RAG pipeline.

1. Phase 1: The Initial Proof-of-Concept

- **Corpus & Scope:** The application originally launched using just two primary core documents—**PH25-I-2** (Citation and Order Writing Handbook) and **PH26-V-1** (Electrical Inspection Procedures Handbook).
- **Performance:** With a tightly bounded document size, a standard fixed-size chunking strategy combined with basic dense vector lookups ran instantly, providing high retrieval accuracy for general compliance questions.

2. Phase 2: The Scaling Wall (Corpus Expansion & Failure Modes)

- **Expansion:** To increase domain coverage, the architecture was aggressively scaled to ingest **MSHA regulatory files and historical sensor telemetry reading handbooks**, ballooning the data footprint to **over 1.5 GB of unstructured and semi-structured text**. This generated nearly **500,000 individual chunks**.
- **How It Failed:**
    - *Retrieval Degradation:* Pure dense vector similarity search suffered from semantic drift. General-purpose embeddings struggled to isolate exact alphanumeric sensor IDs, specific voltage thresholds, and fine-grained citation codes across half a million chunks.
    - *Latency & Memory Bottlenecks:* Indexing and querying 500k chunks standardly overwhelmed local memory limits, causing severe search bottlenecks and high query latencies.
    - *Context Pollution:* Without precise filtering or hybrid scoring, irrelevant telemetry snippets leaked into the LLM context window, triggering hallucinations and breaking the zero-hallucination compliance guardrails.

3. Phase 3: The Production Architecture (Current Model)

To resolve these scaling limitations, the system was refactored into the robust production design active today:

- **Hybrid Search via RRF:** Implemented a dual-index architecture pairing **Chroma (dense vector store using 384-dimension `text-embedding-3-small`)** with **BM25 (sparse keyword index)**, combined via **Reciprocal Rank Fusion (RRF)**. This ensures that exact regulatory codes and equipment identifiers are matched via keyword lexical search while broader conceptual intent is handled via semantics.
- **Optimized Chunk Boundaries:** Standardized to precise **512-token fixed windows with controlled overlap**, optimizing context granularity without sacrificing local paragraph flow.
- **Strict Guardrails & Caching:** Integrated low-temperature generation (`temperature=0.1`) with `gpt-4o-mini`, persistent Streamlit caching, and transparent metadata inspection tools to guarantee enterprise-grade reliability and verifiable source attribution.